

Historical partial note: reversible/canonical strong DF- p -LSI obstruction

Run `fa_banach.001`, lane 2

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Superseded status. This is the intermediate partial result that preceded the full packet in the parent folder. It is retained only as proof history. The current claimed result is the full theorem in `../../main.tex`.

Source question. Bardet–Rouze, arXiv:1803.05379, ask for the range of p 's for which a non-primitive QMS can satisfy a strong inequality $\text{LSI}_{p,\mathcal{N}}(c, 0)$.

this raises the question of finding the range of p 's for which one can find a non-primitive semigroup for which $\text{LSI}_{p,\mathcal{N}}(c, 0)$ holds for some $c > 0$. Moreover, such a no-go result implies the impossibility for any primitive QMS to satisfy $\text{LSI}_{2,\text{CB}}(c, 0)$, as opposed to $\text{LSI}_{1,\text{CB}}(c, 0)$.

All these results rely heavily on the structure and the properties of the amalgamated $\mathbb{L}_p$ spaces. Further development will require better understanding of these spaces, in particular as interpolating Banach spaces.

Historical Partial Theorem

Theorem 1. *Let $1 < p < \infty$. Let $(P_t)_{t \geq 0}$ be a finite-dimensional reversible quantum Markov semigroup whose generator $\mathcal{L}$ has nontrivial fixed algebra $\mathcal{N}$. Then no finite constant c satisfies*

$$\text{Ent}_{p,\mathcal{N}}(X) \leq c \mathcal{E}_{p,\mathcal{L}}(X) \quad (X > 0).$$

In particular, the canonical decoherence generator $\mathcal{L}_{\mathcal{N}} = E_{\mathcal{N}} - \text{id}$ has zero strong p -log-Sobolev constant for every finite $p > 1$.

Idea of the Proof

The construction is the two-level perturbation used in the source paper for $p = 2$. Choose density matrices $\rho_{\mathcal{N},k} \in E_{\mathcal{N}*}(\mathcal{D}(\mathcal{H}))$ and an off-diagonal self-adjoint perturbation $\Delta \in \ker E_{\mathcal{N}*}$ joining eigenvectors with eigenvalues

$$x_k = \lambda_1/k, \quad y_k = (1 - 1/k)\lambda_2.$$

For $\rho_{k,\epsilon} = \rho_{\mathcal{N},k} + \epsilon\Delta$, set

$$X_{k,\epsilon} = \Gamma_{\sigma_{\text{tr}}}^{-1/p}(\rho_{k,\epsilon}^{1/p}).$$

The entropy identity of Bardet–Rouze gives

$$\text{Ent}_{p,\mathcal{N}}(X_{k,\epsilon}) = \frac{\epsilon^2}{p} \frac{\log x_k - \log y_k}{x_k - y_k} + O_k(\epsilon^3),$$

up to a fixed nonzero matrix-entry factor. The displayed divided difference diverges as $k \rightarrow \infty$.

On the other hand, the p -Dirichlet expansion contains only divided differences of $t^{1/p}$ and $t^{1/q}$, where $q = p/(p-1)$. These remain bounded as $x_k \downarrow 0$ and $y_k \rightarrow \lambda_2 > 0$. Reversibility cancels the fixed-algebra second-order terms, leaving a uniformly bounded energy coefficient. Hence

$$\lim_{k \rightarrow \infty} \limsup_{\epsilon \downarrow 0} \frac{\mathcal{E}_{p,\mathcal{L}}(X_{k,\epsilon})}{\text{Ent}_{p,\mathcal{N}}(X_{k,\epsilon})} = 0,$$

contradicting any finite strong p -LSI constant.

How This Was Upgraded

The parent full packet observes that the reversible cancellation can be replaced by a fixed-algebra Hamiltonian cancellation: on the decoherence-free algebra, the generator is a derivation, and the singular projected second-order terms commute with the relevant zeroth-order function of $\rho_{\mathcal{N},k}$. Traciality of σ_{tr} on $\mathcal{N}$ then kills those terms without assuming reversibility.